

Vessel Log On — Functional Specification

Version: 0.5 (draft)

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Status: For operational review; not an approved operating procedure

Domain: Marine rescue vessel log on, watch, and log off

Audience: Anyone implementing or evaluating a system that performs this function

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1. Purpose and scope

1.1 Purpose

This document specifies the required behaviour of a system that records vessels departing under the watch of a marine rescue unit, monitors their safe return, and escalates when they do not return.

It is written to be implementation-independent. It names no product and assumes no technology. It is equally usable as a build specification and as an evaluation checklist for an existing system.

1.2 The primary safety objective

All requirements in this document derive from one objective:

PSO. If a vessel does not return when it said it would, the unit shall know, and the unit shall be able to find it.

PSO expresses the intended outcome; software cannot guarantee that a vessel will be found. Its testable contribution is to preserve the information received, expose missing or uncertain information, detect missed obligations, and retain an accountable watch owner.

Conflicting requirements shall be resolved explicitly during review, with the decision recorded against PSO. Implementers shall not silently waive a requirement by invoking PSO.

1.3 Scope

In scope: capture of log on details across all channels; identification and verification of vessels and persons; search across unit records; monitoring of active log ons; overdue detection and escalation; log off; amendment; record retention and audit.

Out of scope: conduct of the search or rescue itself; asset and crew management; member administration and billing; incident management beyond the point of escalation.

1.4 Priority

Requirements carry a priority reflecting current operational need, not importance:

Priority	Meaning
P1	Primary focus. Capture and verification of radio log ons — §5, §6, §7.
P2	Required, but adequately served by existing practice — §8, §9.
P3	Required for completeness; lowest urgency.

P2 means lower change urgency in the existing operation, not an optional release gate. A replacement capture system shall demonstrate a functioning path into the existing watch or implement the required watch behavior before it is relied on operationally. A saved record alone is not evidence that anyone is monitoring it.

1.5 Requirement language

Shall — mandatory. **Should** — strongly recommended; deviation requires justification. **May** — optional.

2. Definitions

Term	Definition
Log on	A record that a vessel has departed under the unit's watch, with a stated intention to return by a stated time.
Log off	The act of closing a log on because the vessel has returned or otherwise ended its trip.
Operator	A person, usually a volunteer, receiving log ons at a unit.
Unit	A marine rescue base holding the watch for a geographic area.
Member	A person with a standing record held by the organisation.
Public user	A person without a standing record, logging on as a non-member.
Identifier	Any value that resolves to a stored record: member number, vessel registration, mobile number, vessel name.
Verification	Evidence that independently supplied identifiers consistently identify a stored person/vessel association; not proof of the caller or trip facts.
Cross-verification	Comparison of independently captured identifiers and their candidate associations. See §7.
Draft	Capture is incomplete. This does not disable watch monitoring.
Active	An open record in the watch queue, including pending acceptance and unresolved deadlines.
Overdue	At least one open timed obligation has passed its due time without satisfaction or explicit amendment.
Escalated	An open escalation exists. This is independent of capture completeness and deadline amendment.
ETA	A supplied expected return or report date/time, represented by a timed obligation.
Obligation	An expected return, position report, crossing completion, or operator follow-up, with its own status and deadline.
Watch owner	The unit accountable for the open record, including unresolved capture and pending transfer.
Acknowledgment	An operator records seeing an alert or accepting a transfer; neither action implies the vessel has returned.
POB	Persons on board.
Enrichment	Addition of detail to an existing log on after its initial creation.
Airtime	Occupancy of a shared radio channel. A finite, contended resource.

3. Domain model

3.1 Entities

Entity	Description	Key attributes
LogOn	One trip record, with independent capture and watch states.	capture status, watch status, watch owner, channel, operator, call time, departure point/time, destination, POB, verification outcome, created/updated timestamps
Vessel	A boat known to the system.	registration, name, length, hull colour, type, make, model, AIS identifier
Member / PublicUser	A person, with membership a standing-record attribute rather than a prerequisite to capture.	identifiers, name, contact numbers, associated vessels
OnboardContact	A reachable person aboard or ashore; location/role explicitly recorded.	name, aboard/ashore, relationship, number, source and confirmation time
Identifier	An independently captured value and its resolution history.	type, raw value, normalized value, source, captured time, candidates, match basis, selected record, outcome
Obligation	One expected event or internal follow-up.	kind, due date/time, raw time expression, interpretation basis/timezone, status, satisfaction/amendment evidence
EscalationStep	An action under the unit's approved escalation procedure.	procedure version, action, timestamp, actor, outcome, disposition
WatchTransfer	A request and acknowledgment of a change in owning unit.	source/target, requester, requested time, record version, accepting operator/time, outcome
Location	A departure/destination/report point.	name, optional coordinates, free-text flag, source
Operator	The authenticated person taking an action.	identity, unit, shift

This is a conceptual model; it does not prescribe separate database tables for each row. Automated events record a system actor and responsible unit, rather than inventing an operator.

3.2 Relationships

- A LogOn may reference one selected Vessel and one primary Member or PublicUser; either may remain unknown even after capture is marked complete. Additional people are contacts. Vessel/person selection does not require creating a standing record.
- A LogOn retains zero or more captured Identifiers, contacts, obligations, and escalation steps. It retains every transfer attempt and ownership change.
- Vessels and people have many-to-many associations. A phone number or name need not be unique. Registration/identifier namespaces and validity periods must be respected.
- Stored profile changes do not rewrite historical trip evidence. A log on retains the values, source and verification basis used at the time, alongside links to current profiles.

3.3 Capture, watch and escalation are separate

Capture status: Draft or Complete. An operator may mark capture complete with missing or inconsistent information (CAP-12); the gaps remain visible. Complete is not a claim that the vessel is identified, the record is correct, or the watch is established.

Watch status: Pending acceptance, Watching, Logged off, or Cancelled. Every new record is immediately owned by the capturing unit and appears in its open watch queue, including an empty Draft. When shared creation is unavailable, CAP-19–20 require an explicit local-only/unsaved state and the approved operational fallback; a local capture is not falsely presented as present in the shared queue. Pending acceptance means responsibility for resolving the record has been assigned; it does not claim that a completed log on was acknowledged to the vessel. Acceptance is an explicit operator action and may occur before capture is complete.

Deadline condition: No usable vessel deadline, Not yet due, Approaching, or Overdue. Conditions derive from open obligations, independently of Draft/Complete and pending acceptance. A usable deadline on any open record is monitored immediately. Entering a past deadline raises the overdue condition immediately; missing POB or identity does not inhibit it. Missing or uninterpretable times create a conspicuous unresolved condition and operator follow-up, not a fabricated vessel ETA (WAT-9).

Escalation status: None, Open, or Closed. An open escalation remains open after an ETA change until an authorized operator records its disposition. Alert acknowledgment, record completion, and transfer do not close it.

Action	Preconditions and result
Begin capture	Create Draft + Pending acceptance; assign the capturing unit; add to the shared open queue.
Accept watch	Pending acceptance → Watching; record operator/time. Missing information remains visible.
Complete/reopen capture	Change Draft/Complete only; monitoring and ownership do not change.
Deadline passes	Any open record becomes Overdue for the affected obligation; alert under WAT-3.
Amend obligation	Retain previous deadline and reason/source; recompute deadline condition. Existing escalation requires explicit disposition.
Begin escalation	Record the approved procedure and actor. May occur before a deadline where that procedure authorizes it.
Log off	Explicit operator action on an open record, with contact/evidence and time. Resolve each open obligation and escalation explicitly in the closing action; do not silently mark all as fulfilled.
Cancel	Explicitly establish no trip/watch was required (for example no departure, accidental entry, or duplicate); record reason and any canonical record link. Overdue/escalated records require authorized disposition, not deletion.
Request transfer	Keep source ownership and monitoring; record transfer pending.
Accept transfer	Change owner to receiving unit through the acknowledged transfer protocol (WAT-8); preserve all open conditions.
Correct mistaken closure	Authorized reopen with reason, restoring ownership and evaluating unsatisfied deadlines immediately; preserve the closure event.

Closed records remain searchable and auditable. Corrections are appended with an explicit reason; they do not silently resume or cancel monitoring. There is no terminal “Transferred” trip state: transfer changes ownership, not whether the trip is open.

3.4 Log on variants and obligations

Variant	Obligations
Standard	Expected return, plus any agreed follow-up.
Long term	Individually tracked position-report deadlines and, where known, expected return. A report does not log off the trip.
Bar crossing	Its own crossing-completion deadline, optionally within the same trip as a later return deadline. Completing the crossing does not satisfy the return obligation.

Variants use the same trip model. Each missed obligation remains independently visible; fulfilling one does not fulfill another. Report schedules, tolerance and escalation policy must be confirmed by the unit before operational use (Appendix D). Internal operator follow-up deadlines are labeled as such and are never presented as caller-supplied ETAs.

4. Operating context

Context that constrains the requirements. Stated as fact, not as complaint.

OC-1. Log ons arrive by marine radio, telephone, in person, and self-service. Radio is reported as dominant at the observed unit and drives the capture design. Other channels require their own acceptance checks; self-service submissions require a defined unit acceptance/acknowledgment path and shall not imply an established watch merely by submitting.

OC-2. A radio log on is an interview, not a dictation. The caller states a rough intention; the operator elicits the remaining detail by asking. Field order is arbitrary and varies per call.

OC-3. Local observations report varying caller information and operator interview order. The system shall not assume a consistently followed script. This does not establish that no organisational procedure exists; applicable procedures must be identified and represented accurately, particularly for escalation and handover.

OC-4. Airtime is contended. Every question the operator asks occupies a shared channel that other vessels are waiting on. The system shall treat operator questions as a cost to be minimised.

OC-5. Local observations report inconsistent phonetic use and recurring letter confusions. Letters and digits can both be misheard. Appendix B is an initial candidate matching heuristic to validate against observed errors, not a guarantee of identity or a substitute for clarification and approved radio procedures.

OC-6. Identifying detail — vessel registration, mobile number — typically arrives late in a call or only when asked, and sometimes on a later call. It shall not be required early.

OC-7. A requirement for a computer record has been reported locally. Its legislative, policy or contractual basis, permitted media, retention period and evidentiary requirements are **not yet established by this specification**. Appendix D question 5 must be resolved before claiming compliance. Immediate durable capture is justified operationally regardless.

OC-8. Calls arrive in bursts. Peak load is a period of good weather, which is also when the greatest number of vessels are at sea.

5. Requirements — Capture (P1)

CAP-1. The system **shall** persist a log on record containing any subset of fields, including an empty subset, from the moment capture begins. Shared persistence requires the acknowledgment in CAP-19; during an outage CAP-20 governs recovery, and the interface shall not imply that an unacknowledged capture is shared or monitored. *Rationale: retaining partial information prevents a missing operational record. Legal sufficiency remains subject to OC-7. Persistence and failure semantics are defined in §5.2.*

CAP-2. The system **shall not** require any field to be populated in order to create, save, or retain a log on. *Rationale: any mandatory field is a point at which capture can fail entirely (PSO).*

CAP-3. The system **shall** accept field entry in any order, with every field reachable and editable from the keyboard without pointing-device interaction. *Rationale: OC-2. Entry order is dictated by the caller.*

CAP-4. Unsupplied trip facts **shall** remain empty. The system **shall not** invent POB, destination, departure time or ETA. It **may** record authenticated operator, unit, entry time and reliably known channel as system metadata, visibly distinct from reported facts. Previously stored details **shall** be offered with source/age and confirmation status; applying them shall retain that provenance, not imply fresh caller confirmation. *Rationale: a known system fact is not a guessed trip fact. Reuse must not conceal uncertainty.*

CAP-5. Where the system detects an inconsistency — for example an ETA preceding the departure time — it **shall** present the warning adjacent to the field concerned, at the time of entry, and **shall** permit the record to be saved regardless. *Rationale: inconsistency is information; preventing the save discards the record to protect the field.*

CAP-6. The system **shall** permit multiple log ons to be in Draft simultaneously, and **shall** permit an operator to suspend and resume capture of any Draft without loss. *Rationale: a priority radio call may interrupt capture at any point.*

CAP-7. The shared open watch queue, including pending acceptance, unresolved drafts, and overdue log ons, **shall** remain visible during capture of a new log on. *Rationale: situational awareness is the operator's primary function; capture must not suspend it.*

CAP-8. The system **shall** accept time input in the forms operators speak and write, including 24-hour (1500), 12-hour (3pm), and relative (+2h). *Rationale: format conversion is cognitive load during a timed task.*

CAP-9. The system **shall** accept free text for departure point and destination, and **shall** offer known locations as suggestions without restricting entry to them. *Rationale: callers name places that exist in local usage and in no list.*

CAP-10. The system **shall** display, during capture, which fields remain unpopulated. *Rationale: OC-3. In the absence of a procedure, the visible gap list is the procedure.*

CAP-11. The system **shall** rank unpopulated fields by the priority in §6.1, giving greatest prominence to fields serving PSO and least to vessel description. *Rationale: OC-4. Undifferentiated prompting spends airtime on low-value fields.*

CAP-12. Prompting under CAP-10 and CAP-11 **shall** be advisory, and **shall not** prevent an operator from completing or closing a capture. *Rationale: the operator, not the system, judges what is worth a transmission.*

CAP-13. The system **shall** permit any field of an existing log on in any open state to be populated or amended, with the same interaction cost as initial capture. *Rationale: OC-6. Late-arriving detail is the normal case.*

CAP-14. Capture of a log on for a person without a standing record **shall** proceed without first classifying the caller, and **shall not** require vessel description fields before the log on is created. *Rationale: classification is a decision the operator cannot make until identity is known, which is late (OC-6).*

CAP-15. Where vessel or contact detail has been captured previously for the same vessel or person, the system **shall** offer it rather than requesting it again. *Rationale: OC-4. Re-asking known detail spends airspace for no information gain.*

5.1 Capture performance

CAP-16. For a caller whose identifiers resolve to an existing record, and who supplies all requested detail without repetition, a complete log on **shall** meet the **30-second 95th-percentile target**, keyboard only, under CAP-18.

CAP-17. For a caller with no existing record, a complete log on **shall** meet the **90-second 95th-percentile target** under CAP-18.

CAP-18. Simulation **shall** be used before operational trials; it is not sufficient for final performance acceptance. Final evaluation **shall** observe authorized real-call use under peak load without compromising the existing watch. The approved measurement protocol shall define a complete capture, timing start/end, operator experience, sample size, peak-load conditions and interruptions. Report median, 95th percentile, maximum, error/correction rate and live-entry proportion separately for known and unknown callers. The 30/90-second targets apply to the 95th percentile of eligible calls; no field may be fabricated or warning hidden to meet them. Record excluded/interrupted calls separately. Report both end-to-end elapsed capture time and operator interaction time; do not count only keystrokes while excluding normal caller speech from the elapsed measure.

5.2 Persistence, interruption and concurrent work

CAP-19. Begin capture **shall** immediately attempt durable creation and subsequent edits **shall** be retained automatically without a final Save dependency. The interface shall distinguish Saving, Saved to the shared record, and Not saved to the shared record. It shall claim Saved only after durable acknowledgment. Any local recovery copy must be labeled separately; local storage alone does not establish a shared watch.

CAP-20. Loss of connectivity, rejected saves and unavailable shared monitoring **shall** be conspicuous. Captured edits shall be recoverable after interrupted browser sessions on the same supported workstation, subject to the approved recovery envelope. Recovery shall preserve raw input, original capture time and pending actions. Cross-workstation availability shall be stated honestly. The operational fallback and recovery envelope (including workstation/power loss) require approval before live use; no claim of zero loss may be made without demonstrated coverage.

CAP-21. Retrying creation, edits or queued recovery actions **shall not** duplicate trips, obligations, transfer acceptance or closure events. Reconnection shall reconcile pending work without silently overwriting later shared edits. Suspected duplicate calls shall be shown for operator review; uncertain identities shall not trigger automatic merging.

CAP-22. Concurrent changes **shall** preserve both operators' evidence. Conflicting changes, especially ETA, identity, ownership and closure, shall be presented for explicit resolution; stale edits shall not silently replace current values. Operators shall see that another operator is working on the same open record.

CAP-23. Unknown, explicitly unavailable, not applicable, and inconsistent values **shall** remain distinguishable. Unparsed or implausible input shall be retained as captured with a warning rather than silently discarded, rounded or converted into a credible-looking operational value. Only a usable, explicitly interpreted deadline can drive vessel overdue detection; unresolved time input invokes WAT-9.

6. Requirements — Data (P1)

6.1 Field priority classes

Class	Purpose	Fields
A	Locate the vessel	ETA, POB, destination, departure point, departure time
B	Identify and verify	Member number, vessel registration, mobile number
C	Reach the vessel	Radio channel monitored, onboard/shore contact, AIS identifier
D	Describe the vessel	Length, hull colour, type, make, model

DAT-1. Class A fields **shall** be the preferred trip-information set, not a save or watch-acceptance gate. Incomplete records may still be useful. A record without a usable vessel deadline shall be conspicuously marked as not time-monitorable for vessel return, while remaining owned, visible and subject to an operator follow-up deadline (WAT-9).

DAT-2. The mobile number **shall** be treated as both a Class B identifier and a Class C contact, and prioritised accordingly. *Rationale: it serves double duty — it identifies the caller and is the first action on overdue.*

DAT-3. Class D fields **shall not** be requested during capture where the vessel resolves to an existing record carrying them.

DAT-4. The system **shall** retain Class D detail captured for a person without a standing record, and associate it with the vessel for reuse.

DAT-5. Every captured identifier **shall** be stored as captured, independently of whatever record it resolves to. *Rationale: the value the operator heard is evidence; overwriting it with the resolved value destroys the ability to detect a mis-resolution later.*

7. Requirements — Identification, verification and search (P1)

7.1 Cross-verification

IDV-1. The system **shall** support two or more independently supplied identifiers. Repeated entry of the same identifier, a normalized duplicate, or a value populated from the first match shall not count as independent corroboration. Record whether each value was supplied on this call, imported, selected from a profile, or corrected after clarification.

IDV-2. Resolution **shall** operate over compatible person/vessel associations, with explicit handling of non-unique identifiers. Outcomes shall be:

Outcome	Condition
Verified	At least two independent supplied identifiers have exact normalized matches whose intersection uniquely identifies one person/vessel association; no captured identifying evidence contradicts it or remains unresolved.
Conflict	Exact resolving evidence has incompatible candidate associations, or retained evidence explicitly contradicts the selected association.
Partial	Some exact evidence resolves but other evidence is unmatched or only approximate, or more than one compatible association remains. Display the unresolved/ambiguous reason.
Unverified	Fewer than two independent identifiers, or no exact evidence resolves.

Conflict takes precedence over other outcomes. A member with several vessels is not a conflict merely because there are several associations. Shared numbers and names do not prove uniqueness. A vessel identified without a person may be selected and watched, but does not meet the stated Verified person/vessel criterion. The display shall explain which person, vessel and evidence are matched rather than show a badge alone.

IDV-3. Conflict **shall** be shown prominently immediately. The operator shall be able to inspect all captured values and match bases, request clarification, append a correction, select a provisional association with a reason, or leave identity unresolved. A provisional selection shall not dismiss the conflict or relabel it Verified. Explicitly superseded incorrect values remain in history; verification is recomputed from current evidence.

IDV-4. The outcome, evidence, match method and selected association **shall** be stored and shown on the open watch queue and record. Changes shall be versioned. Later profile edits shall not silently change what was verified at the time of the call.

IDV-5. An outcome other than Verified **shall not** block creation, saving, completion, or watch acceptance. Verified refers only to consistency with stored identity associations; it does not authenticate the caller, validate trip facts, confirm contact reachability, or establish a vessel's current location.

7.2 Tolerant resolution

IDV-6. Resolution **shall** offer labeled near-match candidates in addition to exact matches, using Appendix B as an initial heuristic. Exact candidates shall be ranked first; matched/substituted characters and match basis shall be visible. Selecting a near match shall not make the raw captured value exact or establish Verified. Clarification may append a corrected supplied value while retaining the original. *Rationale: tolerate plausible capture errors while keeping uncertainty visible (OC-5).*

IDV-7. Resolution results **shall** include sufficient descriptive detail — vessel name, length, hull colour, type, associated member — to allow the operator to confirm identity by description. *Rationale: descriptive detail offers another route to clarification. Its reliability relative to readback is not established by this specification.*

IDV-8. Multiple candidates **shall** be presented in one comparison view, without silently selecting the first. Large sets may be paginated with visible counts and no hidden truncation. Profile freshness and uncertainty shall be visible.

IDV-9. Normalization rules and identifier namespaces **shall** be explicit and tested. Formatting normalization may yield an exact match; character substitution is approximate. Neither normalization nor profile selection shall overwrite the value as captured.

7.3 Search

SRCH-1. The system **shall** provide a single search input that returns results across all record types: members, public users, vessels, active log ons, draft log ons, and historical log ons. *Rationale: cross-verification (IDV-2) requires two identifiers resolvable in one action. Separate searches per record type make verification cost more airtime than it saves, and it is then not performed.*

SRCH-2. Search **shall** accept any identifier type in the same input without the user selecting a type in advance: member number, vessel registration, mobile number, vessel name, person name.

SRCH-3. A vessel or person **shall** remain returnable by search regardless of whether they currently hold an active log on. *Rationale: an in-progress trip is the state in which lookup is most often needed.*

SRCH-4. Search results **shall** indicate the record type and current state of each result, and **shall** indicate whether a returned vessel or person is currently logged on.

SRCH-5. Search **shall** return results on partial input.

SRCH-6. Every applicable search result **shall** offer one action to apply its identity or offered profile detail to the current capture. Show what will be applied. A historical trip shall not supply current ETA, POB, destination or departure facts; selecting an open trip shall offer resumption rather than silently create a duplicate. Permission boundaries shall apply across all search types (REC-7).

8. Requirements — Watch and overdue (P2)

WAT-1. The system **shall** display the persistent open watch queue, including Drafts, pending acceptance, missing/unusable vessel deadlines, overdue obligations and open escalations. Overdue/escalated and unresolved records shall not be hidden by an ETA-only sort; records with usable deadlines shall show the next deadline and all missed obligations. Age, watch owner, capture status and synchronization health shall remain visible.

WAT-2. The system **shall** indicate log ons approaching their ETA before that ETA passes.

WAT-3. A missed obligation **shall** produce an alert independent of the operator observing a list change. Deadline evaluation shall continue without an open capture page or browser. The approved deployment shall define detection/delivery latency, notification recipients, acknowledgment, repeat/unacknowledged-alert behavior and monitoring-health failure notification. On restart or reconnect, already-missed obligations shall be reconciled and surfaced, not postponed until another state transition. Acknowledgment records attention; it neither satisfies the obligation nor closes escalation.

WAT-4. The system **shall** present the owning unit's approved escalation procedure, including its version, and record attempted actions with time, actor, outcome and any authorized deviation. The procedure, responsibilities and contact paths must be approved before live use. Software shall not invent rescue policy from a generic checklist.

WAT-5. On transition to Overdue, the system **shall** surface the log on's verification outcome (IDV-2). *Rationale: personnel about to act on the record need to know which parts of it are verified.*

WAT-6. Shift handover **shall** present and acknowledge all open records, including incomplete/unresolved capture, pending saves, missed obligations, pending transfers and open escalation. Record outgoing/incoming operators and outstanding responsibilities without re-keying. Unit ownership and monitoring shall persist through shift change.

WAT-7. A record **shall** leave the owning unit's open queue only through explicit, authorized log off, cancellation or acknowledged transfer under §3.3. Record actor, time, reason/evidence and disposition of obligations/escalation. Alerts, refreshes, time passage and changes to capture completeness shall not silently

remove it.

WAT-8. Transfer **shall** preserve the complete record without re-keying. The source unit remains responsible until the receiving unit explicitly accepts the current transfer and durably receives the current open obligations/escalation state. Acceptance shall make one authoritative ownership change, visible to both units, without a monitoring gap. Stale acceptance shall require review of intervening changes. Rejected, timed-out, retried or disconnected transfer attempts shall retain source responsibility until reconciled; the transfer itself shall never fulfill a deadline or close escalation.

WAT-9. Every open record without a usable vessel deadline, including an empty Draft, **shall** have an accountable operator follow-up under the unit's approved policy. The policy shall define the follow-up interval, recipient, reminder and escalation for an unresolved capture, including operator interruption or shift end. Label it as an internal follow-up, not an inferred ETA. Draft status shall never exempt a supplied usable vessel deadline from monitoring.

WAT-10. Each obligation **shall** retain its own fulfillment/amendment evidence. A later report or revised ETA shall not erase a previous miss. An ETA change following escalation shall retain the open escalation until explicit authorized disposition. A new due time shall be shown with its source and the previous due time accessible.

9. Requirements — Record and audit (P2)

REC-1. The system **shall** support durable records, retention, integrity and export. The responsible organisation shall identify and approve the applicable legal/policy basis, retention schedule, access, disposal and export obligations before claiming the system is its compliant official record (OC-7). This draft does not itself establish compliance.

REC-2. Every log on and subsequent action **shall** record actor, unit, event/entry time and available channel/source. Automated events shall identify the system actor; unknown channels remain unknown. Reported call time and departure time shall remain separate from automatic entry time. Late entries shall retain both event and entry times.

REC-3. Amendments **shall** be appended. Prior values **shall** remain retrievable. *Rationale: post-incident review requires knowing what was known at each point in time, not only the final state.*

REC-4. Verification outcomes and raw identifiers (DAT-5) **shall** remain part of the audit record for the approved retention period, including superseded interpretations.

REC-5. The system **shall** report on operational measures including call volume by period, overdue frequency, escalation outcomes, and **the proportion of log ons captured during the call versus entered afterwards**. *Rationale: the latter measures whether capture is keeping pace with the radio. The measurement protocol shall establish call start/end or a recorded live/late/unknown classification; entry timestamps alone cannot establish when the call occurred.*

REC-6. Interpreted times **shall** include full date, time and timezone. Preserve raw time input and the reference instant for relative expressions. +2h shall use the displayed, recorded reference (normally the call time when known, otherwise the entry time), not be recalculated on reload. Ambiguous day rollover, past-date entry and timezone interpretation shall require clarification or remain unresolved; never silently shift an elapsed deadline to tomorrow. Changes append a new interpretation. Time-source health and acceptable clock error shall be defined in the approved monitoring configuration.

REC-7. Access **shall** be authenticated and authorized by role/unit for search, edit, export, closure, reopening, transfer and escalation. Record access-relevant administrative actions. Session expiry shall not silently lose capture or stop shared monitoring. Public self-service shall not expose other people's records or grant operator privileges.

REC-8. Recovery **shall** be demonstrated by restoring representative records, identifiers, histories, ownership and obligations into an isolated environment. The approved recovery-time and data-loss limits shall be documented, including reconciliation of changes made during outage. Restored missed deadlines shall be evaluated before resuming live watch; a backup file existing is not evidence that recovery works.

10. Acceptance criteria

Each criterion is a pass/fail test against a running system.

10.1 Capture

#	Test	Pass condition
AC-1	Begin a log on, enter nothing, save.	Record persists and is retrievable.
AC-2	Enter only POB and destination. Save. Close the session. Reopen.	Record persists with those two fields.
AC-3	Inspect a new capture and apply offered profile details.	Trip facts remain empty; automatic metadata and reused/unconfirmed profile values are visibly distinguished and retain provenance.
AC-4	Enter an ETA earlier than the departure time. Save.	Warning shown beside the field; save succeeds.
AC-5	Begin a log on, start a second, return to the first.	Both retained, neither altered.
AC-6	Capture a log on while the active list is on screen.	Active list remains visible throughout.
AC-7	Complete a full log on using keyboard only.	Achievable; no pointing device required.
AC-8	Enter 1500, 3pm, +2h; include midnight rollover and late entry.	Raw values retained; full interpreted date/time, timezone and reference shown; ambiguity is clarified or visibly unresolved; reload does not move a relative deadline.
AC-9	Enter a destination not present in any list.	Accepted as entered.
AC-10	Measure eligible known-caller captures under the approved peak-load protocol.	95th percentile $\leq$ 30 seconds; sample, maximum, errors and exclusions reported (CAP-16, CAP-18).
AC-11	Measure eligible unknown-caller captures under the approved peak-load protocol.	95th percentile $\leq$ 90 seconds with the same reporting (CAP-17, CAP-18).
AC-12	Add a registration to a log on created an hour earlier.	Same interaction cost as initial capture.

10.2 Verification and search

#	Test	Pass condition
AC-13	Independently enter exact identifiers uniquely identifying one person/vessel association.	Verified with evidence shown. Duplicate/autofilled evidence does not independently corroborate.

#	Test	Pass condition
AC-14	Enter exact identifiers with disjoint person/vessel candidate associations.	Outcome shown as Conflict , prominently; shared-vessel associations alone do not produce a false conflict.
AC-15	Enter a registration one character wrong within Appendix B; select a suggested match.	Correct vessel offered with substitution marked; selection alone never becomes exact or Verified.
AC-16	Inspect any resolution result.	Includes vessel name, length, colour, type, associated member.
AC-17	Search a registration for a vessel currently logged on.	Vessel returned, marked as currently logged on.
AC-18	Search a member number, a registration, a mobile number and a vessel name in the same input.	All return results without changing search mode.
AC-19	Save a log on with a Conflict outcome.	Save succeeds; conflict remains visible on the record.
AC-20	View the active watch list.	Verification outcome visible per row.

10.3 Watch, record and audit

#	Test	Pass condition
AC-21	Allow an ETA to pass.	Overdue alert raised without operator action.
AC-22	Trigger an overdue on an unverified record.	Verification state surfaced with the alert.
AC-23	Amend an ETA, then inspect history.	Both original and amended values retrievable.
AC-24	Perform a shift handover.	All active log ons and escalation history transfer without re-entry.
AC-25	Run the operational report.	Proportion of live-captured versus later-entered log ons is reported.

10.4 Adoption and release acceptance

AC-26. During an approved operational trial, routine double entry into a parallel paper log ceases because the system supports the work, rather than because operators are told to stop. Measure duplicate-entry proportion and delay against the baseline. Approved outage/contingency records are evaluated separately.

This is an adoption outcome, not a sufficient safety test. Passing it cannot substitute for deadline, persistence, recovery and handover acceptance. Failure requires investigation of workflow, policy and trust rather than assuming operator resistance.

10.5 Interrupted and exceptional operation

#	Test	Pass condition
AC-27	Capture ETA/POB/destination, leave Draft and pending acceptance, then pass ETA.	Record stays owned and visible; overdue alert occurs despite incomplete capture.
AC-28	Begin an empty Draft; interrupt operator until the approved follow-up deadline passes.	Unresolved record has an owner; internal follow-up alerts without fabricating a vessel ETA.

#	Test	Pass condition
AC-29	Fail a save, disconnect, close/reopen the browser, reconnect and retry.	Status never falsely claims shared persistence; recovery meets the approved envelope; no duplicate trip or silent loss/overwrite.
AC-30	Two operators amend ETA or identity from the same version; one closes the record while the other is editing.	Stale/conflicting changes require explicit resolution; both evidence histories remain; closure is not silently undone.
AC-31	Supply shared phone numbers, a member with several vessels, unmatched evidence and incompatible exact matches.	Outcomes follow IDV-2; ambiguity is explained; conflict takes precedence; raw inputs remain.
AC-32	Request transfer, pass ETA before acceptance, retry after disconnect, then accept after an intervening edit.	Source remains responsible until current-state acceptance; alert is delivered without a gap; one ownership change; stale state cannot silently be accepted.
AC-33	Miss a position report on a long-term trip with a later return ETA; complete a nested bar crossing.	Missed report remains visible/alerted; crossing completion does not fulfill report or return obligations.
AC-34	Change ETA on an escalated trip and acknowledge its alert.	Old miss and actions remain; escalation closes only through explicit authorized disposition.
AC-35	Close all browsers and pass a deadline; restart monitoring with already-missed obligations; leave an alert unacknowledged.	Detection, delivery, health failure and repeat behavior meet approved limits; acknowledgment does not log off a vessel.
AC-36	Log off, cancel a duplicate, and correct a mistaken closure with appropriate roles.	Reasons/evidence retained; obligations/escalations receive explicit disposition; reopen evaluates missed deadlines; unauthorized actions refused.
AC-37	Expire a session, attempt cross-unit/private search/export without permission, and submit through self-service.	Capture recovery and shared monitoring persist; access boundaries enforced; submission does not falsely acknowledge watch acceptance.
AC-38	Restore a backup and reconcile outage records in isolation.	REC-8 recovery limits met; audit and ownership preserved; pending/missed obligations recovered without duplicate actions.
AC-39	Change a vessel/person profile after a completed call.	Historical identifiers, applied details and verification evidence remain as known at the call; current profile changes are distinguishable.

10.6 Operational release gate

Before operational reliance, the unit shall approve the unresolved policy/configuration items in Appendix D that affect this deployment, and record a traceability checklist for all applicable requirements. In particular, define measurable alert latency, clock error, follow-up intervals, escalation/transfer authority, recovery limits and trial metrics. These values are not invented by this functional specification.

All applicable acceptance criteria shall pass on the intended deployment, including its existing-watch integration if P2 functions are retained elsewhere. Simulation and isolated failure/recovery tests precede a supervised live trial with continuity of the established watch. Unsupported variants/channels shall be explicitly declared and excluded from the initial rollout; they shall not appear available without their monitoring behavior.

Appendix A — Justification from current practice

Evidence that the requirements above address real conditions rather than hypothetical ones. Recorded from a unit operating a commercial resilience-management platform alongside a handwritten radio log.

A.1 The paper log is the working record. Its columns are: date/time of call; member number or vessel registration; contact mobile; POB; departure point; destination; return date; return time; *entered into the computer system*; *system record ID*; logged off.

The presence of the last-but-two and last-but-one columns is decisive. **The paper log tracks the computer system as an outstanding task.** The operational record is paper; the computer record is a downstream transcription. The computer record therefore lags the operational one; its legal status remains subject to confirmation under OC-7. AC-26 exists to detect when this has been reversed.

A.2 All-or-nothing saving produces the backlog. The current system cannot persist an incomplete log on. An operator who has captured half the detail has nothing to save, so the entry stays on paper until there is time to complete it — and the backlog grows fastest precisely under peak load (OC-8), when it is least affordable. CAP-1 and CAP-2 address this directly.

A.3 Mandatory fields are the fields that arrive last. Registration and mobile number are required before a record can be saved, and are the two fields that typically arrive last in a call or on a later call (OC-6). The form demands first what the conversation surrenders last. CAP-2, CAP-13 and CAP-14 address this.

A.4 Default values that fail validation. The observed new-log-on form pre-populates departure time and ETA with the same value, then rejects the record on the grounds that the ETA must be later than the departure. The form's initial state fails its own validation, so every log on begins from an error condition. If saved unnoticed, the paired defaults would record a zero-duration trip and could produce immediate or premature overdue handling. The screenshots do not establish how the existing overdue engine would behave; that requires a runtime check. CAP-4 and CAP-5 address this.

A.5 Validation errors are remote from their fields. Errors are presented as a block at the top of the form, referring to fields several sections away, discovered only on attempting to save. CAP-5 addresses this.

A.6 Capture occludes the watch. New log on capture opens as a full-screen modal, concealing the list of vessels currently at sea for its duration. CAP-7 addresses this.

A.7 Search is partitioned, and cross-verification is thereby defeated. Members, vessels and active log ons are held behind separate searches. A vessel holding an active log on is not returned by vessel search — the state in which lookup is most often needed. Consequently, cross-verifying two identifiers requires several lookups across separate screens, and under load it is not performed. The unit's principal accuracy mechanism therefore stops operating exactly when accuracy matters most. SRCH-1 to SRCH-4 address this.

A.8 Two identifiers are the existing accuracy practice. Operators routinely request two identifying values — member number and registration, or registration and mobile number. Agreement provides corroboration; disagreement signals a possible error. This is not a guarantee of identity, and shared, stale or correlated identifiers require the ambiguity rules in §7. The practice is undocumented and unsupported by software, and is performed by eye across the partitioned searches described in A.7. IDV-1 to IDV-5 formalise it.

A.9 Phonetic discipline is not reliable under load. Requesting a phonetic repeat costs a transmission on a contended channel. Local observations report that repeats are sometimes omitted under load. IDV-6 supports recovery from plausible mishearing; it does not justify abandoning approved radio procedures or necessary clarification.

Appendix B — Letter confusion set

The following initial groups reflect reported confusion patterns, not validated error probabilities. IDV-6 shall use them to suggest candidates, subject to field-specific identifier formats and namespaces. They shall never establish an exact match.

Group	Members
Rhyming ("E-set")	B · C · D · E · G · P · T · V · Z · 3
Nasal	M · N
Sibilant	F · S · X · 6
Long-A	A · J · K · 8
Letter/digit collision	0 ↔ O · 1 ↔ I ↔ L · 5 ↔ S · 2 ↔ Z

Digits shown in the table participate in those candidate substitutions. Other digit errors remain possible and shall be preserved for clarification rather than presumed reliable. Evaluate candidate recall and false suggestions against representative calls; clarify local pronunciations (including Z) and identifier formats before tuning rankings.

Appendix C — Traceability from version 0.3

0.3	0.4
§1 safety contract	§1.2 PSO
§1.2 legal record	OC-7, REC-1
§2.2 interview	OC-2
§2.3 no procedure	OC-3
§2.4 phonetics	OC-5, Appendix B
§3.1 tiers	§6.1 classes A–D
§3.2 checksum	IDV-1 ... IDV-5
§3.3 unified search	SRCH-1 ... SRCH-6
C1, C2	CAP-1, CAP-2
C3	CAP-4
C4	CAP-5
C5	CAP-6
C6	CAP-7
C7	CAP-16 ... CAP-18
C8	CAP-8, CAP-9
C9, C10	CAP-10, CAP-11, CAP-12
C11	IDV-6 ... IDV-8

0.3	0.4
C12	IDV-1 ... IDV-5
C13	CAP-13
C14	SRCH-1 ... SRCH-6
W1 ... W7	WAT-1 ... WAT-8
R1 ... R5	REC-1 ... REC-5
§7 anti-requirements	Dissolved into positive requirements; evidence moved to Appendix A
§8 acceptance test	AC-26

Appendix D — Open questions

1. On a **Conflict** outcome (IDV-2), what clarification does the operator perform today? Validate the IDV-3 workflow against that practice and define acceptable evidence for correction or an explicitly unresolved/provisional association.
2. Is a logged-on vessel excluded from vessel search by query construction or by data model? Determines the difficulty of SRCH-3.
3. What is the typical transcription backlog — count of entries, and elapsed time from call to computer entry? Establishes the baseline for REC-5 and AC-26.
4. Which identifier pair is most commonly used: member number with registration, or registration with mobile number? Determines what IDV-2 optimises for.
5. What legal, policy or contractual instrument underlies the reported computer-record obligation (OC-7)? Identify the authority, permitted record media, retention period and integrity obligations under REC-1; do not presume a legislative basis.
6. For long term log ons (§3.4), what reporting schedule is expected, and what constitutes a missed report?
7. Which approved procedures govern escalation, early escalation, alert acknowledgment, closure/reopening and handover? Who may authorize deviations and which version applies?
8. Who owns pending acceptance and unresolved drafts? What internal follow-up intervals, recipients and overdue/unacknowledged-alert behavior should apply at each operating period?
9. What detection/delivery latency, clock-error bounds, monitoring-health notification, recovery-time and data-loss limits can the intended deployment demonstrate?
10. How do source and receiving units accept transfers, including after hours, refusal, connectivity loss and changes while acceptance is pending?
11. What identifier namespaces, shared-number patterns, stale records and person/vessel associations occur? What clarification evidence may supersede a conflicting raw value?
12. What sampling protocol establishes live-call timing and transcription delay? Define peak load, sample size, start/end, complete capture, interruptions and privacy handling.
13. Which channels and variants are included at initial release? What existing watch integration remains authoritative, and how will its acknowledgment/failure be shown?
14. What access, retention, disposal, export and contingency-record policies apply? Identify the document/authority and approver for each release-affecting answer.

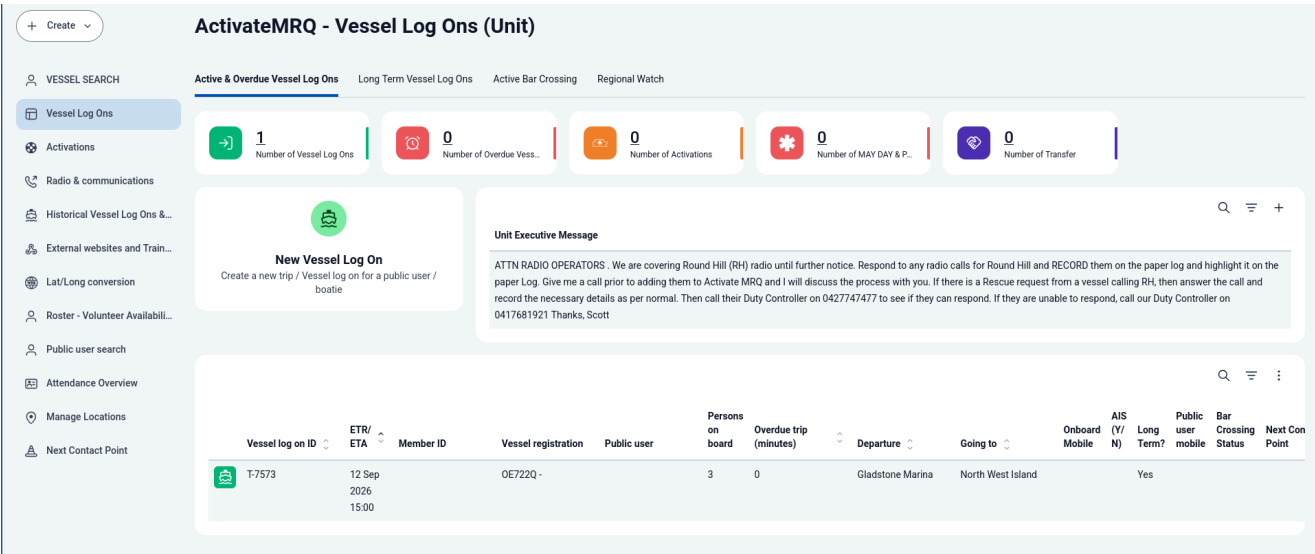
Appendix E — Changes from version 0.4

- Separated capture completeness, watch acceptance/ownership, deadline condition and escalation. Drafts with usable deadlines are monitored; unresolved records get internal follow-up. Transfers retain ownership until explicit current-state acceptance.
- Represented return, position-report and crossing deadlines as independent obligations.
- Added persistence acknowledgment, recovery, retry, concurrent-edit and duplicate handling.
- Distinguished system metadata, supplied facts and reused profile data; preserved time expressions, interpretation basis, call time and entry time.
- Defined independent evidence, ambiguous associations, conflict precedence and approximate candidates. Verified no longer implies caller authentication or confirmed trip facts.
- Added approved alert/escalation configuration, access control and demonstrated recovery.
- Corrected the unsupported equal-time/never-overdue claim and qualified unverified legal, procedural and phonetic assertions. Adoption is now separate from safety acceptance.
- Retained requirement IDs, revised their wording where necessary, and added CAP-19–23, IDV-9, WAT-9–10, REC-6–8 and AC-27–39. CAP-16/17 targets are interpreted by revised CAP-18.
- Added a linked table of contents and included the existing evidence figures in the formatted specification. The figures contain operational/contact data; sharing beyond the authorized audience requires an appropriately redacted copy.

Appendix F — Evidence figures

These are supplied screenshots of the existing platform, not proposed interface designs. Static images evidence visible layout/defaults; search exclusions, persistence behavior, and overdue behavior require observations or runtime checks beyond a screenshot.

Figure 1 — Active log on list



Current platform: active log on list and navigation

Supports the visible dashboard/navigation observations in A.6–A.7. Does not independently prove that an active vessel is excluded from a search query.

Figure 2 — Validation error block

Cancel

New Vessel Log On

Save

Vessel Log On details (mandatory) *

Other details *

Departure point should be entered

Destination date and time should be greater than current date and time

Destination date and time should be greater than departure date and time

Going to field should be entered

Communication method

Radio

>

* Radio channel

82

>

↺

* Is the vessel log on for a known vessel or existing member?

Yes

>

Known Public User

* Vessel

Search by public user name, phone, membership ID, vessel registration number or name

>

Onboard contacts

+

Add an item

Current platform: validation messages separated from fields

Supports A.3–A.5: required controls and errors remote from affected fields.

Figure 3 — Capture form scrolled

Cancel

New Vessel Log On

Save

▼ Known Public User

* Vessel

Search by public user name, phone, membership ID, vessel registration number or name

>

Onboard contacts

+

Add an item

▼ Trip Details

* Number of persons on board

▼

* Departure date and time

12 Sep 2026 06:31

📅

Departure point (list of popular locations)

>

Nearest MRQ unit to the departure location

>

* Estimated date and time of arrival/return at destination

12 Sep 2026 06:31

📅

* Returning to the same location as departure?

Yes

>

Going to

>

AIS Tracking?

Select your option

>

Additional Information

Current platform: scrolled capture form and trip details

Supports A.4 and A.6: trip fields/defaults and capture occupying the screen.

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Figure 4 — Trip-detail defaults

Cancel

New Vessel Log On

Save

Add an item

▼ Trip Details

* Number of persons on board

* Departure date and time

12 Sep 2026 06:31

Departure point (list of popular locations)

Nearest MRQ unit to the departure location

* Estimated date and time of arrival/return at destination

12 Sep 2026 06:31

* Returning to the same location as departure?

Yes

Going to

AIS Tracking?

Select your option

Additional Information

¶ B I U 🔥 ☰ ☷ ☹ ☹ ☹ ☹ ☹ ☹

Enter text here...

Current platform: departure and return fields with identical defaults

Supports A.4: identical displayed departure/return values. No claim about the actual runtime overdue calculation can be established from this image alone.

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